

# Image Analysis Report

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## Executive Summary

This inspection identifies high-severity corrosion concentrated on the structural anchor bolts and nuts, which poses a significant risk to the base's load-transfer integrity. While the galvanized base plate shows mostly surface staining, the fasteners exhibit advanced oxidation and likely material loss.

## Key Findings

| # | Finding                                                                                                                                   |
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| 1 | Heavy atmospheric corrosion and thick rust scaling on all visible anchor bolt threads and double-nut assemblies.                          |
| 2 | Potential section loss on fastener shanks, indicating a High severity rating for critical structural components.                          |
| 3 | Extensive rust staining/bleeding on the galvanized base plate and vertical stiffeners, likely caused by iron oxide runoff from the bolts. |
| 4 | Accumulation of soil and organic debris at the base-to-foundation interface, which promotes moisture retention.                           |
| 5 | Vertical stiffener welds appear structurally intact but are obscured by oxide staining.                                                   |
| 6 | The condition of the anchor bolt shank below the nut level and within the concrete cannot be determined from this image.                  |
| 7 | The original material specification and grade of the bolts cannot be determined visually.                                                 |

## Recommendations

| # | Recommendation                                                                                                                         |
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| 1 | Perform mechanical cleaning (SSPC-SP 3 power tool cleaning) on all fasteners to remove scale for a detailed thickness assessment.      |
| 2 | Conduct a dimensional survey to measure the remaining bolt diameter and compare against original design specifications.                |
| 3 | Replace all heavily corroded nuts and bolts if cross-sectional area loss exceeds 10% of the nominal diameter.                          |
| 4 | Apply a high-solids epoxy coating or cold galvanizing compound to the cleaned base plate and stiffeners to arrest further degradation. |
| 5 | Install protective, grease-filled bolt caps over the anchor assemblies to provide a long-term barrier against atmospheric moisture.    |
| 6 | Clear all vegetation and debris from the base perimeter to ensure proper drainage and reduce the time-of-wetness.                      |
| 7 | Perform a torque test on all anchor bolts to verify that the structural connection still meets required tension specifications.        |

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| 8 | Schedule a follow-up NDT (Ultrasonic Testing) of the anchor shanks to check for subsurface corrosion or cracking. |
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